

Swapnil Maharana

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Berhampur, Odisha

Profile

M.Tech Computer Science student specializing in AI-driven healthcare systems, full-stack development, and federated learning. Experienced in building scalable deep learning applications with real-time deployment using FastAPI and React. Strong background in computer vision, machine learning, and system design.

Technical Skills

- **Languages:** Python, Java, C
- **Web & Backend:** React, HTML, FastAPI
- **AI/ML:** Deep Learning (CNN, DenseNet121, ResNet), Computer Vision (OpenCV, Grad-CAM), NLP (Rasa), Federated Learning (Flower, FedAvg)
- **Tools:** PyTorch, SQLite, Git, Render
- **Core Concepts:** Data Structures & Algorithms, OOP

Projects

FedMed: Privacy-Aware Multi-Disease Chest X-Ray Classification System

- Developed and deployed a production-ready full-stack AI healthcare system for multi-disease chest X-ray classification using DenseNet121 (PyTorch)
- Designed and deployed FastAPI backend and React frontend for real-time inference
- Achieved 87.6% accuracy (F1-score: 86.9%, AUC: 0.91)
- Designed privacy-aware architecture with in-memory processing and secure authentication (Bcrypt)
- Implemented interactive dashboard, analytics, and PDF report generation
- Integrated SQLite database for user management and prediction tracking
- Designed federated learning-ready architecture using Flower and FedAvg
- **GitHub:** github.com/Swapnil4228/FedMed_Render

Smart Farmer Assistant: AI Multilingual Agricultural Decision Support System

- Built AI system for crop yield prediction, disease classification, and crop recommendation
- Achieved $R^2 = 0.9626$ (yield prediction) and 93.26% accuracy (classification)
- Developed multilingual chatbot (English, Hindi, Telugu) using Rasa NLP framework
- Integrated real-time APIs (weather and market data)
- Designed profit-based recommendation system combining yield, cost, and market data
- Combined machine learning, NLP, and decision intelligence into a unified system

Human Activity Recognition using Machine Learning

- Built machine learning system for classifying human activities using sensor dataset
- Processed dataset with 7000+ samples and 500+ features
- Implemented Logistic Regression, SVM (Linear & RBF), Decision Tree, Random Forest
- Applied GridSearchCV for hyperparameter tuning and optimization
- Optimized model performance using hyperparameter tuning (GridSearchCV)
- Achieved 96.6% accuracy using Linear SVM (best model)
- Evaluated performance using confusion matrix and classification metrics
- **GitHub:** github.com/Swapnil4228/Human-Activity-Recognition-Prediction

Scene Classification & Explainable AI using Grad-CAM

- Developed CNN-based model with residual blocks for scene classification
- Applied Grad-CAM to generate heatmaps for visual explanation of predictions
- Improved model interpretability and debugging

- Demonstrated application in scene recognition and satellite imagery analysis

Certifications

- Intro to Deep Learning – Kaggle (2025)
- Foundations of Cybersecurity – Google (Coursera)
- Explainable AI (Grad-CAM) – Coursera
- Software Testing – Microsoft & NIIT

Education

Master of Technology (Computer Science)	2024–2026
Gandhi Institute of Technology and Management, Visakhapatnam	

2024

Bachelor of Technology

National Institute of Science and Technology, Berhampur	2020
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Higher Secondary

Council of Higher Secondary Education
The Mother's International HS School

Matriculation

Central Board of Secondary Education Harobino Vidya Bhawan, Berhampur	2018
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Languages

- English
- Hindi
- Odia